SOLAR FARM
SITE ANALYSIS
MARICOPA COUNTY, AZ

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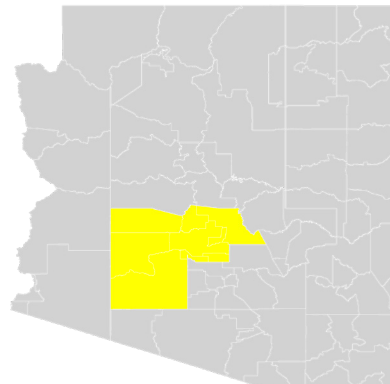
CONTENTS

1 INTRODUCTION

The following report uses GIS to perform a suitability analysis aimed at locating potential sites for a Solar Energy Farm within Maricopa County, Arizona. Solar farms provide an opportunity to harvest clean, renewable energy from the sun using photovoltaic solar panels. The following site analysis uses 7 criteria established using a literature analysis of several research studies performing similar studies. The two literature sources primarily referenced were Choi et al., 2021 and Selim et al., 2021 in which Solar Farm site analyses were performed in Mongolia and Turkey, respectively.

MARICOPA COUNTY



CRITERIA

The following 7 criteria were used to identify suitable conditions for a Solar Farm:

1. Aspect
2. Elevation
3. Slope
4. Distance to Transformers
5. Land Cover
6. Average Annual Temperature
7. Global Horizontal Irradiance (GHI)

RASTER DATA

Maricopa_County_DEM.tif
Annual_GHI_Western-Hemisphere_2020.tif
PRISM_Mean_Annual_Temperature_2020.tif
NLCD_AZ_2016.tif

VECTOR DATA

Maricopa_County_Subdivisions_2018.shp
Maricopa_County_Land_Use_2020.shp
HIFLD_USA_Electric_Substations_2019.shp

METHODOLOGY

The data sources above were used in ArcMap to create the site analysis. Pre-analysis processing included combining the DEM raster files and extracting them to the Maricopa County boundaries using the subdivisions shapefile. Similarly, the remaining raster data files were extracted to the Maricopa County Boundary. Next using select by attributes, the electric substations in Maricopa County exclusively were made into their own layer. The same was done with the land use shapefile; private developable vacant parcels were selected.

Analysis processing included using geoprocessors to obtain the Aspect and Slope. The land cover, GHI and annual temperature data was reclassified using pre-determined cut-offs. Finally, in order to use the raster calculator to perform the final analysis, the substation vector file needed to be converted into the raster format and distance buffers needed to be created and reclassified. This was done using the Euclidean distance tool with a cell size of 10000 and a processing extent set to match the Maricopa_County_DEM.tif layer.

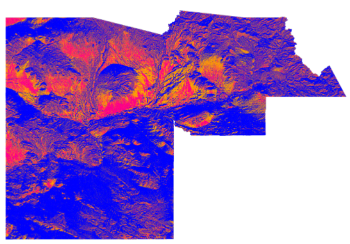
3 CRITERIA

The following 7 criteria were used to identify suitable conditions for a solar farm. Each criteria is mapped below in three classes according to suitability.

HIGH SUITABILITY **MODERATE SUITABILITY** **POOR SUITABILITY**

GEOGRAPHY

FIGURE 5. **ASPECT**



South

Southeast/ Southwest

Others

100

N Kilometers

FIGURE 6. **ELEVATION**

500 - 2331 m

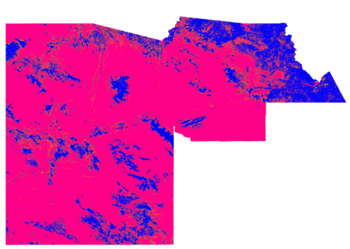
150 - 500 m

Less than 150 m

100

N Kilometers

FIGURE 7. **SLOPE**



Less than 5%

5 - 10%

Greater than 10%

100

N Kilometers

LOGISTICS

FIGURE 1. **DISTANCE TO ELECTRIC TRANSFORMERS**

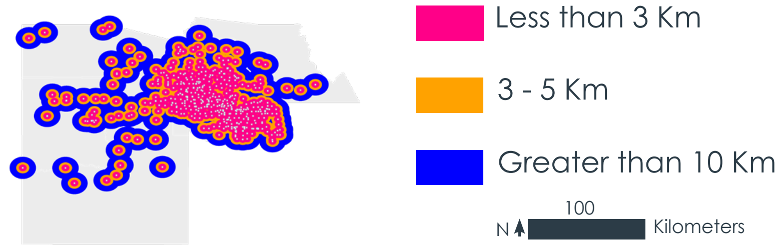
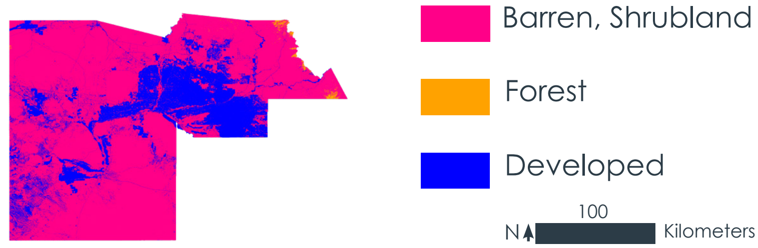


FIGURE 2. **LAND COVER**



SOLAR PARAMETERS

FIGURE 3. **AVERAGE ANNUAL TEMPERATURE**

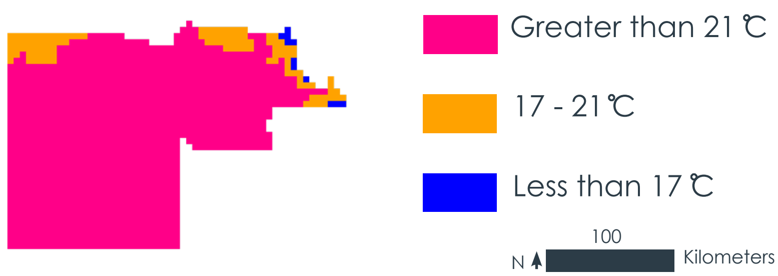
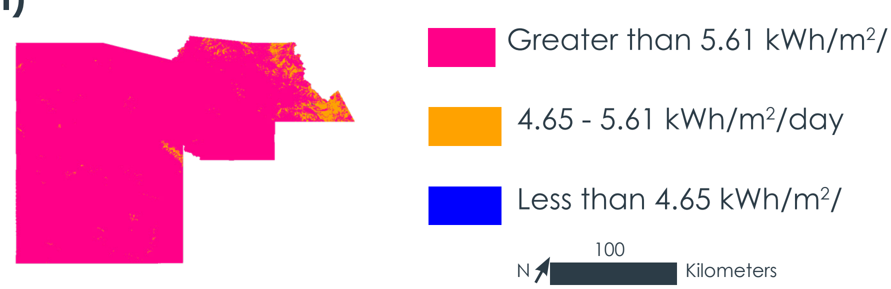


FIGURE 4. **AVERAGE GLOBAL HORIZONTAL IRRADIANCE (GHI)**



5 RESULTS

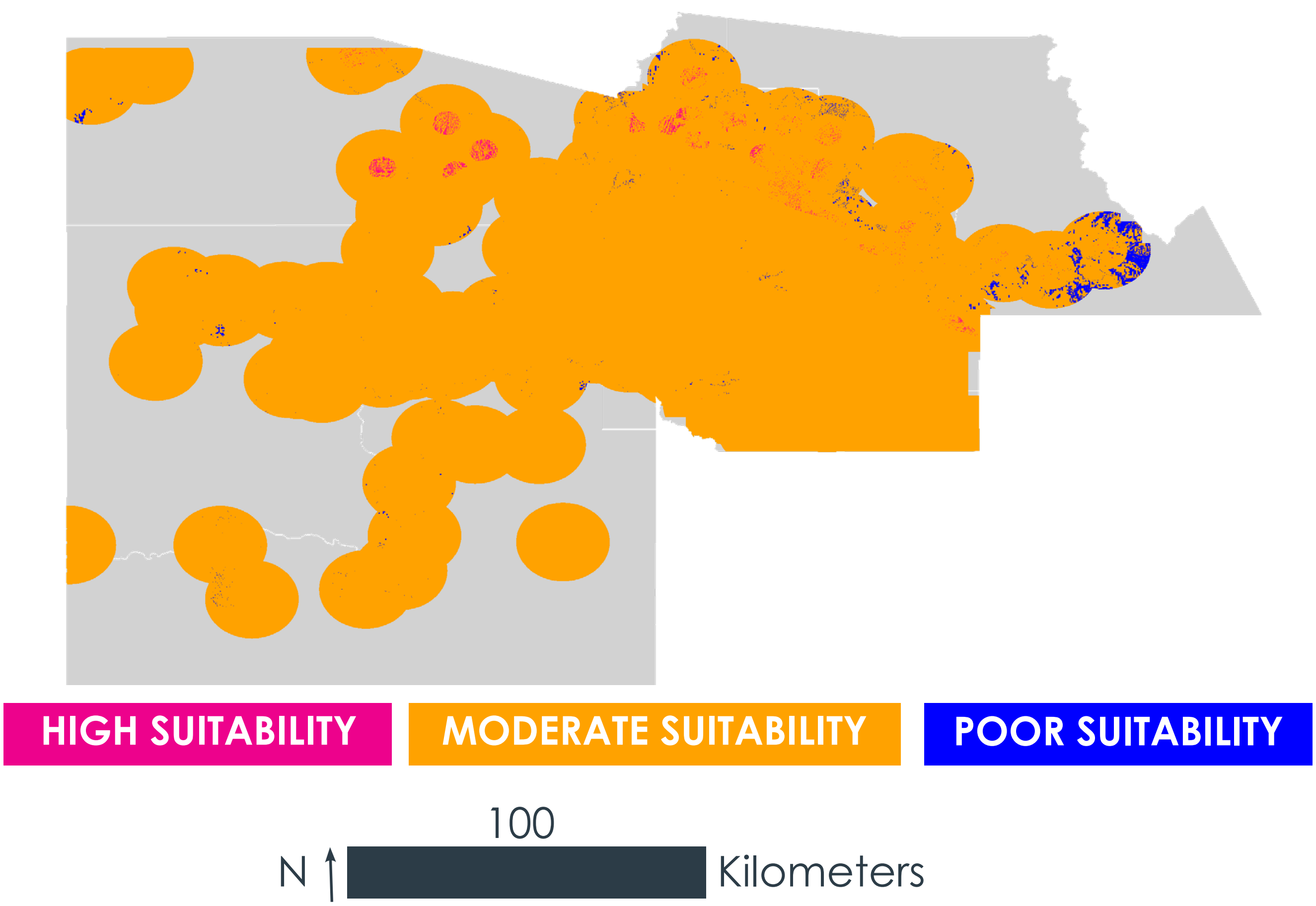
CRITERIA WEIGHTS

Aspect = 10%
Elevation = 5%
Slope = 15%
Distance to Electric Transformers = 20%
Land Cover = 15%
Annual Temperature = 5%
Global Horizontal Irradiance (GHI) = 30%

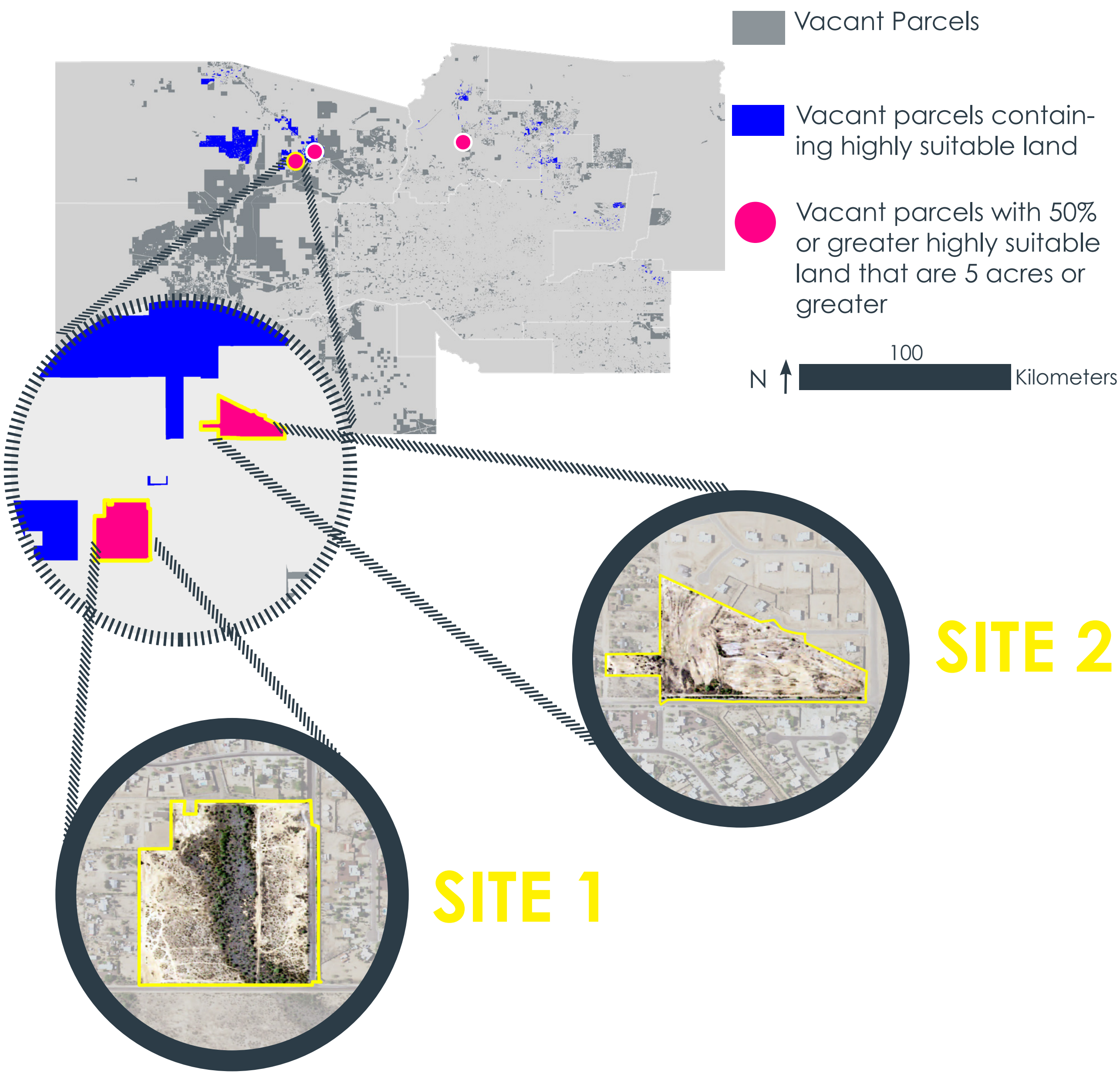
The final suitability surface raster was created using the Raster Calculator in ArcMap. Each of the criteria used were assigned a weighted percentage, based upon their revelance to the final raster.

FUNCTION USED

(aspect_reclass * 10 + elevation_reclass * 5 + slope_reclass * 15 + euclidean_dist_substation_reclass * 20 + land_cover_reclass * 15 + annual_temp_reclass * 5 + GHI_reclalss * 30) / 100



Using Zonal Statistics, the highly suitable land in Maricopa County was cross referenced with vacant parcels in the area. This suitability analysis produced a total of 87.1 acres across 23 vacant parcels in which more than 50% of the land was considered highly suitable. To narrow down the potential sites further, parcel size was taken into account. A total of five parcels were identified as potential sites, being 5 or more acres in size and containing a proportion of 50% or more highly suitable land. Two of these suitable sites are analysed further, below:



SITE SELECTION 6

7 SITE SELECTION

SITE 1: 11.6 mW Solar Farm

Location: Wittmann, AZ
Size: 29.1 Acres
Highly Suitable Land (%): 52.9%
Generated Energy Potential: 1392 MWh/ month

Pros:
- close proximity to residential areas
- accessible from main road

Cons:
- contains shrubs that would require clearing prior to development

SITE 2: 11.6 mW Solar Farm

Location: Wittmann, AZ
Size: 17.4 Acres
Highly Suitable Land (%): 53%
Generated Energy Potential: 804 MWh/ month

Pros:
- close proximity to residential areas
- accessible from main road

DATA SOURCES

https://prism.oregonstate.edu/recent/
https://solargis.com/maps-and-gis-data/download/usa
https://hifld-geoplatform.opendata.arcgis.com/datasets/electric-substations/explore?location=5.689355%2C-9.857173%2C2.56&showTable=true
https://www.arcgis.com/home/item.html?id=0f3f2464210544868499f4f45af889ed
https://azgeo-open-data-agic.hub.arcgis.com/datasets/AZMAG::existing-land-use-for-maricopa-and-pinal-counties-arizona-2020/explore?location=33.244747%2C-111.892947%2C8.83

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